

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Module 2

Network operations

Introduction to network protocols

Welcome to module 2

Video • 1 min

Network protocols

Video • 3 min

Common network protocols

Reading • 4 min

Additional network protocols

Reading • 4 min

Antarix: Working in network security

Video • 2 min

Wireless protocols

Video • 1 min

The evolution of wireless security protocols

Reading • 4 min

Test your knowledge: Recognize different network protocols

Practice Assignment • Grade: 100%

System identification

Firewalls and network security measures

Video • 4 min

Virtual private networks (VPNs)

Video • 2 min

Security zones

Video • 3 min

Subnetting and CIDR

Reading • 4 min

Proxy servers

Video • 3 min

Virtual networks and privacy

Reading • 4 min

VPN protocols: Wireguard and IPsec

Reading • 4 min

Test your knowledge: System identification

Practice Assignment • Grade: 100%

Review: Network operations

Wrap up

Video • 51 sec

Glossary terms from module 2

Reading • 4 min

Module challenge 2

Graded Assignment • Grade: 90%

Module 3

Secure against network intrusions

Introduction to network intrusion tactics

Welcome to module 3

Video • 25 sec

The case for securing networks

Video • 1 min

How intruders compromise your system

Reading • 4 min

Matt: A professional on dealing with attacks

Video • 3 min

Secure networks against Denial of Service (DoS) attacks

Denial of Service (DoS) attacks

Video • 4 min

Read tcpdump logs

Reading • 4 min

Real-life DoS attack

Reading • 4 min

Activity: Analyze network layer communication

Practice Assignment • 36 min

Activity Exemplar: Analyze network layer communication

Reading • 4 min

Test your knowledge: Secure networks against Denial of Service (DoS) attacks

Practice Assignment • 10 min

Network attack tactics and defense

Malicious packet sniffing

Video • 4 min

IP Spoofing

Video • 2 min

Overview of interception tactics

Reading • 8 min

Identify: Network attacks

Ungraded Plugin • 10 min

Activity: Analyze network attacks

Practice Assignment • 10 min

Activity Exemplar: Analyze network attacks

Reading • 4 min

Test your knowledge: Network interception attack tactics

Practice Assignment • 8 min

Review: Secure against network intrusion

Wrap up

Video • 44 sec

Glossary terms from module 3

Reading • 4 min

Module challenge 3

Graded Assignment • 50 min

Module 4

Security hardening

Read tcpdump logs

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A **network protocol analyzer**, sometimes called a packet sniffer or a packet analyzer, is a tool designed to capture and analyze data traffic within a network. They are commonly used as investigative tools to monitor networks and identify suspicious activity. There are a wide variety of network protocol analyzers available, but some of the most common analyzers include:

- SolarWinds NetFlow Traffic Analyzer
- ManageEngine OpManager
- Azure Network Watcher
- Wireshark
- tcpdump

This reading will focus exclusively on tcpdump, though you can apply what you learn here to many of the other network protocol analyzers you'll use as a cybersecurity analyst to defend against any network intrusions. In an upcoming activity, you'll review a tcpdump data traffic log and identify a DoS attack to practice these skills.

tcpdump

tcpdump is a command-line network protocol analyzer. It is popular, lightweight—meaning it uses little memory and has a low CPU usage—and uses the open source libpcap library. tcpdump is text-based, meaning all commands in tcpdump are executed in the terminal. It can also be installed on other Unix-based operating systems, such as macOS[®]. It is preinstalled on many Linux distributions.

tcpdump provides a brief packet analysis and converts key information about network traffic into formats easily read by humans. It prints information about each packet directly into your terminal. tcpdump also displays the source IP address, destination IP addresses, and the port numbers being used in the communications.

Interpreting output

tcpdump prints the output of the command as the sniffed packets in the command line, and optionally to a log file, after a command is executed. The output of a packet capture contains many pieces of important information about the network traffic.

```
20:00:29.528395 IP 198.168.10.1.41 > 198.111.123.1.61012: Flags [P,], seq 120176, ack 1, win 501, options [nop,nop,TS val 4106659748, ecr 2979487360], length 144
```

Some information you receive from a packet capture includes:

- Timestamp:** The output begins with the timestamp, formatted as hours, minutes, seconds, and fractions of a second.
- Source IP:** The packet's origin is provided by its source IP address.
- Source port:** This port number is where the packet originated.
- Destination IP:** The destination IP address is where the packet is being transmitted to.
- Destination port:** This port number is where the packet is being transmitted to.

Note: By default, tcpdump will attempt to resolve host addresses to hostnames. It'll also replace port numbers with commonly associated services that use these ports.

Common uses

tcpdump and other network protocol analyzers are commonly used to capture and view network communications and to collect statistics about the network, such as troubleshooting network performance issues. They can also be used to:

- Establish a baseline for network traffic patterns and network utilization metrics.
- Detect and identify malicious traffic.
- Create customized alerts to send the right notifications when network issues or security threats arise.
- Locate unauthorized instant messaging (IM), traffic, or wireless access points.

However, attackers can also use network protocol analyzers maliciously to gain information about a specific network. For example, attackers can capture data packets that contain sensitive information, such as account usernames and passwords. As a cybersecurity analyst, it's important to understand the purpose and uses of network protocol analyzers.

Key takeaways

Network protocol analyzers, like tcpdump, are common tools that can be used to monitor network traffic patterns and investigate suspicious activity. tcpdump is a command-line network protocol analyzer that is compatible with Linux/Unix and macOS[®]. When you run a tcpdump command, the tool will output packet routing information, like the timestamp, source IP address and port number, and the destination IP address and port number. Unfortunately, attackers can also use network protocol analyzers to capture data packets that contain sensitive information, such as account usernames and passwords.

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Give me a summary

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This reading introduces network protocol analyzers, focusing on tcpdump: as a key tool for cybersecurity analysts.

Understanding Network Protocol Analyzers

- Network protocol analyzers, also known as packet sniffers, capture and analyze network traffic to identify suspicious activity.
- Common examples include SolarWinds NetFlow Traffic Analyzer, ManageEngine OpManager, Azure Network Watcher, Wireshark, and tcpdump.

tcpdump: A Command-Line Tool

- tcpdump is a lightweight, command-line network protocol analyzer that uses the libpcap library and is compatible with Unix-based operating systems.
- It provides brief packet analysis, converting network traffic information into human-readable formats, including source/destination IP addresses and port numbers.

Interpreting Output and Common Uses

- tcpdump output includes a timestamp, source IP and port, and destination IP and port, which are crucial for understanding network communications.
- These tools are used to establish network traffic baselines, detect malicious traffic, troubleshoot performance issues, and identify unauthorized network activity, though attackers can also use them for malicious purposes.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the primary function of tcpdump in network analysis?

How can you use tcpdump to detect a DoS attack in practice?

How might attackers exploit tcpdump to compromise network security?

Bottom

Ask me anything

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https://www.coursera.org/learn/networks-and-network-security/supplement/vqo1C/read-tcpdump-logs

1/1